MICROSCORE

FINANCIAL INCLUSION THROUGH SMARTER SCORING

ENGINEERING CASE STUDY

Engineering a responsible credit-risk system for thin-file borrowers

MicroScore combines interpretable machine learning, a role-based FastAPI product, tenant-aware data infrastructure, and reproducible Monte Carlo stress testing around a financial-inclusion problem in Pavlodar, Kazakhstan.

0.775	152	52 / 52	3
UCI benchmark RF ROC-AUC	automated release-gate tests	PostgreSQL adapter methods	connected engineering layers

THE CORE QUESTION

Can behavioral financial signals help a human loan officer review borrowers who have little formal credit history - without hiding uncertainty, proxy risk, or operational constraints?

The project answers with a working research-and-product system, then documents where the evidence fails. Its most important finding is that the synthetic model is too dependent on repayment-history proxy information for real thin-file claims.

SCAN FOR LIVE REVIEW MODE

https://alex-tereshkovv.github.io/micro-score/#/review

Alexandr | Pavlodar, Kazakhstan | September 2026

01

PROBLEM FRAMING

A local problem, treated as a systems problem

Thin-file borrowers can be rejected because the financial system cannot observe them well, not necessarily because they are unreliable. Alternative data may improve visibility, but it can also encode wealth, digital access, geography, and past access to formal credit. That makes this both an engineering and governance problem.

Product goal	Support a trained MFI analyst with probability, risk band, explanations, lifecycle evidence, policy trade-offs, and portfolio uncertainty.
Safety boundary	The first version does not approve or reject loans automatically. The public demo uses synthetic data and does not collect real borrower identity or banking records.

Four design principles

Interpretability

A score must expose local positive and protective factors.

Traceability

Every decision remains linked to model version, application state, and audit evidence.

Uncertainty

Policy choices are stress-tested as ranges, not presented as guaranteed forecasts.

Claim discipline

Synthetic, public benchmark, and future local evidence are kept separate.

Why Pavlodar

The regional scaffold represents urban, industrial-city, peri-urban, and rural contexts. It is useful for testing workflow and monitoring design, but it is explicitly labeled as simulated context until replaced by measured local evidence.

02

SYSTEM ARCHITECTURE

One traceable path from consent to portfolio evidence

The prototype is deliberately broader than a notebook. It joins a reproducible ML layer to operational API contracts and a reviewer-ready browser experience.

01

Borrower intake

Typed signals, explicit consent, validation, privacy-safe application history.

02

Scoring service

Versioned probability, risk band, warnings, and additive explanation factors.

03

Human review

Analyst packet, affordability context, governance flags, decision history.

04

Policy stress test

Seeded Monte Carlo, paired scenarios, risk appetite, evidence dossier.

Implementation map

Layer	Implementation	Engineering evidence
Research	Python, pandas, scikit-learn	Leakage checks, ablation, calibration, error and segment analysis
Product API	FastAPI, typed schemas, repository boundary	Role access, tenant scoping, immutable score provenance, audit events
Persistence	SQLite prototype + PostgreSQL adapter	52/52 adapter methods and reviewed migration contract
Public demo	Static HTML/CSS/JavaScript	Browser-local synthetic API and GitHub Pages deployment
Verification	unittest + Node smoke workflows	152 tests plus live API, security, research, and frontend gates

Key boundary

The static demo proves the user journey without receiving personal data. The local FastAPI system proves the product contracts. Neither is presented as a production lending deployment.

03

RESEARCH DESIGN

Two experiments, one explicit claim boundary

Experiment A tests a Pavlodar-oriented product and research workflow on synthetic borrower-level data. Experiment B runs the same evaluation discipline on a real public credit-risk benchmark. They answer different questions and are not blended into one claim.

EXPERIMENT A	EXPERIMENT B
Synthetic Pavlodar scaffold	**UCI Default of Credit Card Clients**
5,000 synthetic rows for feature stress tests, product behavior, regional segmentation, and decision-policy research.	30,000 public Taiwan credit-card rows for an external benchmark of ranking, calibration, and error-analysis machinery.
Claim: the system can expose a fragile thin-file signal and support reviewer workflow design.	Claim: the research pipeline operates on a real public dataset. It does not validate Kazakhstan deployment.

Evaluation pipeline

- Leakage-like identifiers and target-adjacent fields removed before the primary baseline.
- Stratified train/test split plus five-fold cross-validation with fixed random state 42.
- Logistic Regression and Random Forest baselines with ROC-AUC, Brier score, F1, precision, and recall.
- Feature-group ablation, single-feature proxy audit, calibration curves, and false-positive / false-negative analysis.
- Three-zone approve / review / decline policy analysis with segment-level outcomes.
- Seeded Monte Carlo portfolio scenarios separated from borrower-level scoring.

Reproducibility contract

Generated artifacts store data label, row count, random state, test size, metric tables, calibration bins, error examples, and chart outputs. The same repository release gate rebuilds and validates the pipeline.

04

QUANTITATIVE EVIDENCE

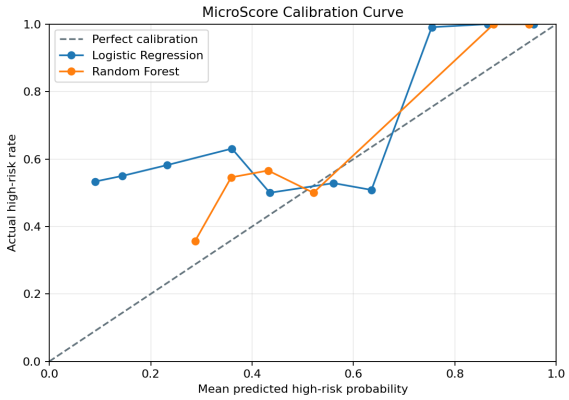
Performance is useful; failure analysis is decisive

The public benchmark shows that the pipeline can rank risk on real public data. The synthetic experiment shows why a strong headline metric must still be challenged by ablation and calibration analysis.

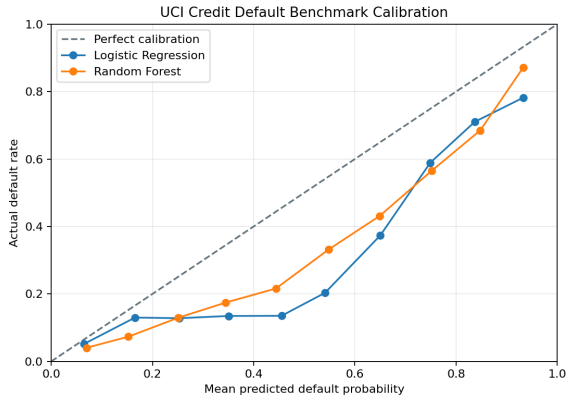
Model comparison

Dataset	Model	ROC-AUC	Brier	F1	Interpretation
Synthetic	Logistic Regression	0.806	0.185	0.793	Interpretable API baseline
Synthetic	Random Forest	0.830	0.143	0.777	Highest synthetic ranking
UCI public	Logistic Regression	0.710	0.209	0.465	Real public benchmark
UCI public	Random Forest	0.775	0.159	0.540	Best benchmark result

Calibration views



Synthetic experiment calibration



UCI public benchmark calibration

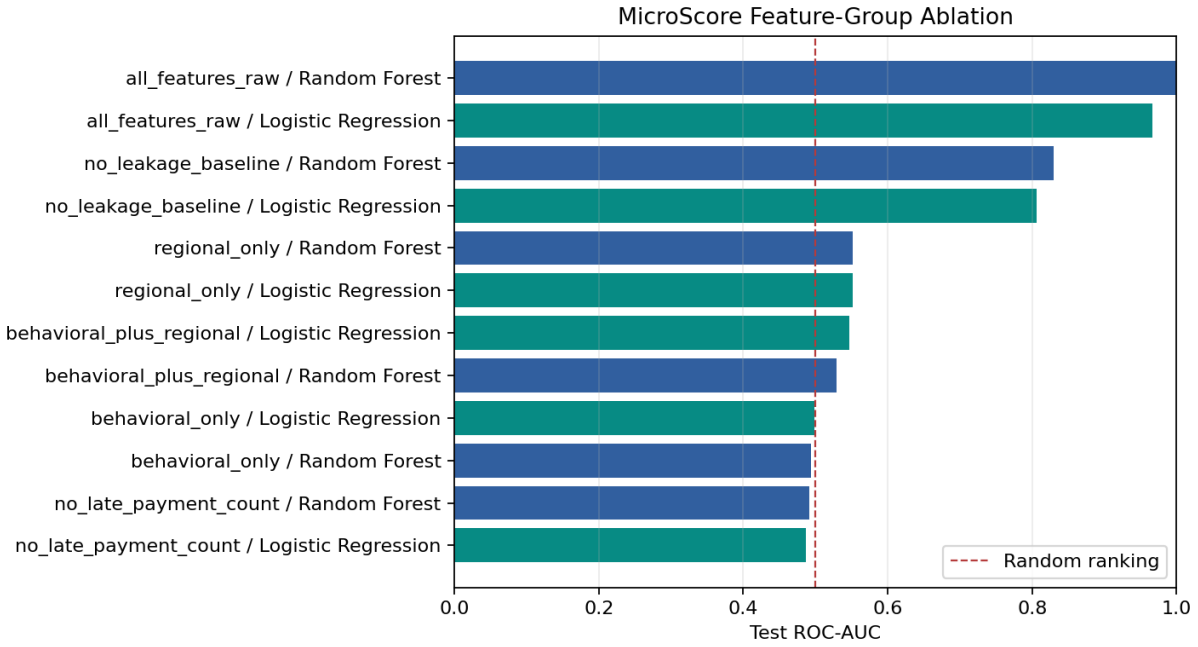
ROC-AUC measures ranking, while the Brier score and calibration curve test whether probabilities behave like probabilities. Both matter because portfolio simulations consume probability estimates, not only rank order.

05

CRITICAL FINDING

Remove one proxy and the thin-file claim collapses

The synthetic Random Forest reaches ROC-AUC 0.830, but falls to 0.492 when late_payment_count is removed. The single feature alone has directional ROC-AUC 0.827. This is the central research result.



0.830	0.492	0.494	0.551
full leakage-safe synthetic RF	RF without late-payment proxy	behavioral-only RF	regional-only RF

Engineering interpretation

The product must not market alternative behavioral data as proven predictive signal. The correct response is to narrow the claim, preserve human review, monitor the proxy, and design a consented local validation plan.

Why this matters

- A high aggregate metric can be produced by a field that contradicts the intended thin-file use case.
- Digital and monetary variables may measure infrastructure and wealth access instead of repayment reliability.
- Honest negative results make the next data requirement concrete and prevent unsafe deployment claims.

06

DECISION ENGINEERING

From a score to policy trade-offs and uncertainty

MicroScore avoids a single binary threshold. A three-zone policy separates auto-approve, manual review, and auto-decline regions, then exposes how each policy changes access and risk. Monte Carlo sits above that policy layer and never changes borrower scores.

Deterministic policy comparison

Policy	Approve	Review	Decline	High-risk approval
Lender Protective	12.7%	26.5%	60.8%	9.3%
Balanced Review	27.8%	24.3%	47.9%	20.6%
Inclusion First	39.2%	27.0%	33.8%	28.8%
Starter Loan Review	23.9%	39.4%	36.7%	17.2%

Monte Carlo uncertainty layer

INPUTS	SIMULATION	OUTPUTS
Scored portfolio Policy thresholds Interest margin Loss given default Operating cost	Seeded paired draws Shared macro shock Borrower-level calibration shock Baseline / adverse / severe Standard-error diagnostics	Approvals and exposure Defaults One-period result distribution Downside and capital buffer Risk appetite gate

Interpretation boundary

These distributions demonstrate a reproducible uncertainty method. They are not forecasts because the borrower probabilities and financial assumptions are not calibrated on local MFI outcomes or verified KZT economics.

Operational outputs

- Scenario cockpit and policy sweep for paired comparisons under the same seed.
- Assumption sensitivity ranking to identify which inputs move downside the most.
- Committee brief, pilot risk appetite gate, monitoring plan, and evidence dossier.
- Device-local dossier vault for reproducible reviewer packets in the public demo.

07

PRODUCT ENGINEERING

Three roles connected by one auditable lifecycle

The product prototype turns research outputs into explicit user responsibilities. Borrowers control submissions, analysts own human review, and administrators inspect governance evidence without exposing raw secrets.

Role	Primary workflow	Safety and evidence
Borrower	Register, consent, submit, view owned status timeline	Borrower-safe response projection; no internal score or analyst note leakage
MFI analyst	Triage queue, inspect explanation, review packet, decide	Lifecycle guards, decision history, model-use notice, tenant-scoped portfolio
Administrator	Manage staff, sessions, model registry, audit and readiness	MFA prototype, invite hygiene, security evidence room, pre-pilot gate

Security and infrastructure evidence

Tenant isolation

Organization-scoped queue, analytics, exports, simulation registry, and analyst lifecycle.

Model governance

Candidate registration, atomic activation, immutable provenance, and stale-score warnings.

Identity controls

Expiring invites, one-time token handling, session inventory and revocation, prototype MFA.

Storage migration

Reviewed PostgreSQL schema and 52/52 repository adapter methods, while production readiness stays blocked.

Readiness is a computed state

The admin pre-pilot gate aggregates security, identity, delivery, storage, model, review-flow, privacy, tenant-isolation, and Monte Carlo evidence. It deliberately keeps production_data_allowed=false while blockers remain.

08

VERIFICATION

The release gate is part of the product

The repository uses a single local and CI release gate. It checks research behavior, API contracts, privacy boundaries, frontend workflows, static-demo parity, storage migration contracts, and live security flows.

152	13	52 / 52	1
Python unit and integration tests	local smoke workflows	PostgreSQL adapter methods	repeatable release command

Release gate coverage

Gate	What it proves
Research smoke	Model training, leakage-safe features, regional analysis, and threshold outputs execute reproducibly
Static demo smoke	Role flows, privacy guards, model registry, Monte Carlo, tenant isolation, and readiness controls remain aligned
Frontend workflow smoke	Borrower history, review checklist, decision mutation guards, and terminal lifecycle behavior
PostgreSQL migration smoke	Schema inventory, JSON mapping, tenant indexes, and repository contract completeness
Live API smoke	Temporary SQLite end-to-end application scoring and decision history
Live security smoke	Invite lifecycle, MFA monitoring, sessions, webhook replay protection, and pre-pilot blockers

Verified release snapshot

The release candidate passed all 152 tests plus research, frontend, static-demo, PostgreSQL migration, live API, security, recovery, observability, and performance smoke checks before publication.

Reproduce locally

```
powershell -ExecutionPolicy Bypass -File scripts\check.ps1
```

09

CONCLUSION

What is proven, what is not, and what comes next

MicroScore demonstrates the engineering of a responsible decision-support system: reproducible research, explicit failure analysis, a typed product API, auditable human review, tenant-aware infrastructure, and uncertainty tooling. It does not demonstrate that the current model is ready to make lending decisions in Kazakhstan.

PROVEN IN THE PROTOTYPE	STILL BLOCKED
- End-to-end role workflows	- Real Pavlodar predictive validation
- Reproducible model evaluation	- Verified KZT financial assumptions
- Public benchmark execution	- Production identity provider and MFA
- Proxy and ablation failure analysis	- External transactional delivery
- Seeded portfolio stress methodology	- Managed PostgreSQL deployment
- Audit and readiness evidence	- Security and privacy review for real data

Next engineering milestone

Run a consented, privacy-reviewed pilot validation with measured local definitions, recalibrate monetary assumptions in KZT, and compare model and reviewer outcomes over time. Until then, preserve the human-in-the-loop boundary.

EXPLORE THE PROJECT

Live engineering review

https://alex-tereshkovv.github.io/micro-score/#/review

Source and reproducibility

https://github.com/alex-tereshkovv/micro-score

Primary evidence sources: generated CSV and PNG artifacts under reports/research-artifacts and reports/benchmark-artifacts; implementation and contracts in the linked repository. UCI source: archive.ics.uci.edu/dataset/350/default+of+credit+card+clients.